

2022 GRADE 11 CHEMISTRY EXAMINATION

MULTIPLE CHOICE QUESTIONS

MEMO

[20 marks]

1.1 B✓✓

1.2 B✓✓

1.3 D✓✓

1.4 C✓✓

1.5 D✓✓

1.6 D✓✓

1.7 C✓✓

1.8 B✓✓

1.9 D✓✓

1.10 B✓✓

QUESTION 2**FM****13 MARKS**

2.1 $n = cV$ ✓ (method) = $2 \times 0,1$ ✓ = **0,2** ✓ mol
(accuracy; no penalty for no SI unit) (3)

2.2 $n(\text{NaOH}) = cV = 0,6 \times 0,04$ ✓ (method) = **0,024** mol
 $n(\text{NaOH}):n(\text{HCl}) = 1:1$ ✓ (method)
Therefore, $n(\text{HCl}) = 0,024$ ✓ mol (accuracy; no penalty for no SI unit) (3)

2.3 $n(\text{HCl}) = 0,2 - 0,024$ ✓ (method) = **0,176** ✓ mol (2)

2.4 $n((\text{NH}_4)\text{O}):n(\text{HCl}) = 1:2$ ✓ (method; accuracy)
Therefore $n((\text{NH}_4)\text{O})$ that reacted = $\frac{1}{2} \times 0,176 \text{ c.o.e}$ ✓ = 0,088 mol
Mass of $(\text{NH}_4)\text{O}$ in 5 g = nM ✓ (method) = $0,088 \times 52$ ✓ = 4,576 g
 $\% \text{ Purity} = \frac{4,576}{5} \times 100\% = 91,52 \%$ ✓ (accuracy) (5)

OR Moles used initially = $\frac{5}{52} = 0,096$

$\% \text{ Purity} = \frac{\text{moles used}}{\text{moles initially}} = \frac{0,088}{0,096} \times 100\% = 91,67 \%$ ✓ (accuracy)

QUESTION 3**FM****20 MARKS**

3.1.1 Explain why fluorine. (4)

- (Both F_2 and Br_2 molecules are non-polar molecules) with London (dispersion) forces between them.✓
- Br_2 molecules have many more electrons✓ in them than F_2 molecules, therefore more London forces can form between their molecules (OR there can be a greater imbalance in electron distribution in Br_2 molecules OR larger temporary dipole)✓
- Therefore the intermolecular forces in bromine are stronger than those in fluorine. More E is required to overcome these forces therefore higher BP✓

3.1.2 Explain why the boiling points of hydrogen bromide (HBr), hydrogen chloride (HCl), and hydrogen fluoride (HF) differ. (5)

- Both HBr has dipole-dipole ✓ (van der Waals) forces between them.(✓)
- HF has hydrogen bonds✓ between its molecules.(✓)
- Hydrogen bonds are stronger than dipole-dipole forces.✓
- Therefore HF requires the greatest amount of energy to overcome its intermolecular forces✓

3.1.3 The boiling points of CH_3F and CH_3Br differ by about $80^\circ C$.

Name the type of intermolecular force that is responsible for the substance named in (a) having the higher boiling point.

Dipole-dipole ✓ forces (2)

3.2

3.2.1 $(K_2SO_4) \rightarrow 2K^{1+} + SO_4^{2-}$ ✓ (2)

3.2.2 Covalent bond✓✓ (NOT ionic bond - intermolecular force) (2)

3.2.2 Ca^{2+} ions have twice the (electrostatic) attraction (force) on sulfate ions than K^+ ions.✓

The ionic bonds between cations and anions in the calcium sulfate lattice are twice as strong as each bond between K^+ and SO_4^{2-} ions.✓
It takes more energy to break these bonds. ✓ (3)

3.2.3 $K_2SO_4(s) \rightarrow 2 K^+(aq) + SO_4^{2-}(aq)$ ✓✓ (2)

(water or H_2O can be written on the arrow)
(arrow in one direction only)
(phase symbols can be left out)

3.2.4 Potassium sulfate (K_2SO_4) ✓✓ (2)

QUESTION 4 **FM****15 MARKS**

- 4.1 the net change of chemical potential energy ✓ of the system ✓ (2)
- 4.2
- 4.2.1 $\Delta H = E_a - E_{out} \checkmark, = 8 - 54.1 = -46.1 \checkmark \checkmark \text{ kcal.mol}^{-1}$ (3)
- 4.2.2 Endothermic curve ✓ ✓ (2)
- 4.2.3 Reaction coordinate ✓ (1)
- 4.2.4 E_a ✓ and ΔH ✓ correctly labelled. (2)
- a) Platinum, palladium, Nickle ✓ **NAME** (1)
- b) The activation energy of the reaction is not lowered. ✓
An alternative reaction with a lower energy is followed ✓. (2)
- c) Lower curve ✓, same position for reaction and products. ✓

QUESTION 5 KH**34 MARKS**

- 5.1.1 $n = m/M \checkmark = 2.54/63.5 \checkmark = 0.04 \text{ mol} \checkmark$ (3)
- 5.1.2 $n = c \times V \checkmark = 0.8 \times 0.1 \checkmark = 0.08 \text{ mol} \checkmark$ (3)
- 5.1.3 0.03 mol $\checkmark$ of Cu is required $\checkmark$ and 0.04 mol $\checkmark$ is available, hence in excess. (3)
- 5.1.4 8:2, $\checkmark$ therefore 0.02 mole of NO
 $V = n \times V_m \checkmark = 0.02 \times 22.4 \checkmark = 0.448 \text{ dm}^3 \checkmark$ (4)
- 5.2
- 5.2.1 average rate = $V/t \checkmark = 0.448 \text{ dm}^3 \checkmark \text{ COE}/28-32 = 0.014 \text{ dm}^3 \cdot \text{s}^{-1} \checkmark$ (3)
- 5.2.2 B – lower gradient $\checkmark$ and half of A $\checkmark$
C – higher gradient $\checkmark$ and double A $\checkmark$ (4)
- 5.2.3 More particle per unit volume $\checkmark$
more collisions per unit time $\checkmark$
More effective collisions (correct orientation and E_k above E_a) $\checkmark$
Hence, an increase in reaction rate. $\checkmark$ (4)
- 5.3
- 5.3.1 DECREASE $\checkmark$, NO is a heavy gas $\checkmark$, the gas particles would escape the vessel $\checkmark$ (3)
- 5.3.2 DECREASE $\checkmark$, the blue colour will intensify as more $3\text{Cu}(\text{NO}_3)_2$ is made $\checkmark$, hence less light would be transmitted. $\checkmark$ (3)
- 5.4
- 5.4.1 a reaction involving the transfer of electrons $\checkmark$ (1)
- 5.4.2 $\text{Cu} \rightarrow \text{Cu}^{2+} \checkmark + 2\text{e}^- \checkmark$ (2)
- 5.4.3 YES $\checkmark \checkmark$ (2)

QUESTION 6 KH

28 MARKS

6.1

6.1.1 reactions which transform thermal energy ✓ into chemical potential energy ✓ (2)

6.1.2 **ENDOTHERMIC** ✓ (1)

6.1.3 **ENDOTHERMIC** ✓ (1)

6.1.4 MORE ✓ (1)

6.1.5 **POSITIVE** ✓✓ (2)

6.2

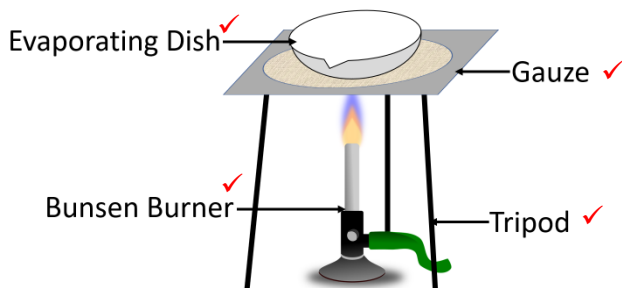
6.2.1 a transfer of electrons ✓ and subsequent electrostatic attraction ✓ (2)

6.2.2 Ammonium drawn for cation ✓
chloride drawn for anion ✓ (4)

Cation	Anion
$\begin{array}{c} \text{H}^+ \\ \vdots \\ \text{H}:\text{N}:\text{H} \\ \vdots \\ \text{H} \end{array}$ ✓	$\begin{array}{c} \cdot\cdot \\ :\text{Cl}-: \\ \cdot\cdot \end{array}$ ✓

6.3 Q

6.3.1



6.3.2 Mass of barium hydroxide = 97,190 - 1,260 = 95,93 ✓
molar mass – 171,3 g.mol⁻¹ ✓
 $n = m/M = 95,93/171,3 = 0,56 \text{ mol}$ ✓

Mole of water = 0,07 ✓

Ratio, 0,56: 0,07 ✓
8:1

Therefore $x = 8$ ✓

(6)

6.4

6.4.1 as the change in concentration per unit time✓ of either a reactant or product
✓ (2)

6.4.2 Y-axis correct (Accept: fraction of particles, % particles, amount (moles) of particles; molecules instead of particles.) ✓

• Shape of higher-temperature graph: peak must be to the right and below the peak of the lower-temperature graph AND stays above the original graph✓

• Activation energy indicated✓

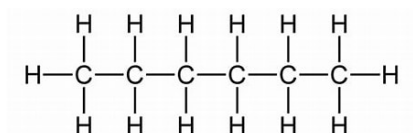
• Proportion of particles indicated✓ (4)

QUESTION 7 SG

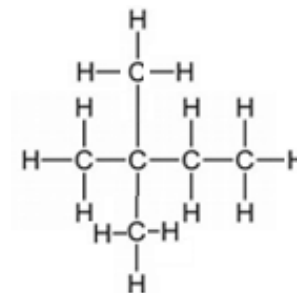
23 MARKS

- 7.1 compounds having the same molecular formula ✓ but different structural formulae ✓ (2)
- 7.2 $C_n H_{2n+2}$ ✓ (2)
- 7.3 C_6H_{14} ✓ ✓ (2)
- 7.4

Hexane

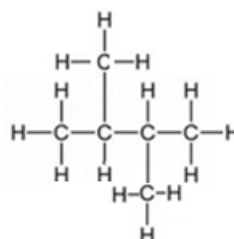


2,2-dimethylbutane



- Correct number of C-atoms in main chains ✓ ✓ ✓ ✓
- Both methyl branches on C-2 in -butane ✓

- 7.5 SATURATED ✓ (1)
- 7.6 hexane ✓ (1)
- 7.7 Hexane has the longer chain therefore greater surface area ✓ than its isomer.
(There are more points of contact in the longer chain.)
Therefore a greater number of London (dispersion) forces between molecules of hexane ✓ makes these the stronger IMF. ✓
(2,2-dimethylbutane is a smaller (more compact) molecule so it has less (weaker) London forces between its molecules.) (3)



- 7.8 Methyl on 2C ✓ and 3C ✓, four carbons ✓
- 7.9
- 7.9.1 Thermal ✓ cracking ✓ (2)
- 7.9.2 Meth ✓ ane ✓ (2)

QUESTION 8 SG**17 MARKS**

- 8.1 1-chloro✓propane✓ (2)
- 8.2 A series of similar compound with the same general formula and functional group✓, where each member differs from the previous by a CH_2 unit. ✓ (2)
- 8.3 Haloalkanes✓ (1)
- 8.4
- 8.4.1 $\text{CH}_3\text{CH}✓\text{CH}_2✓$ (2)
- 8.4.2 $\text{CH}_3\text{CH}_2\text{CH}_2✓\text{OH}✓$ (2)
- 8.5 Elimination✓ (1)
- 8.6 Hydrolysis✓✓ (2)
- 8.7 Dehydration✓✓ (2)
- 8.8 The reactant are volatile and escape the reaction vessel. ✓
The reflux apparatus reintroduces the reactants in to the rxn vessel. ✓ (2)

QUESTION 9 SG**30 MARKS**

9.1 Q

9.1.1 Condensation (esterification)✓ (1)

9.1.2 An atom or group of atoms✓ that form the centre of chemical activity of a compound. ✓ (2)

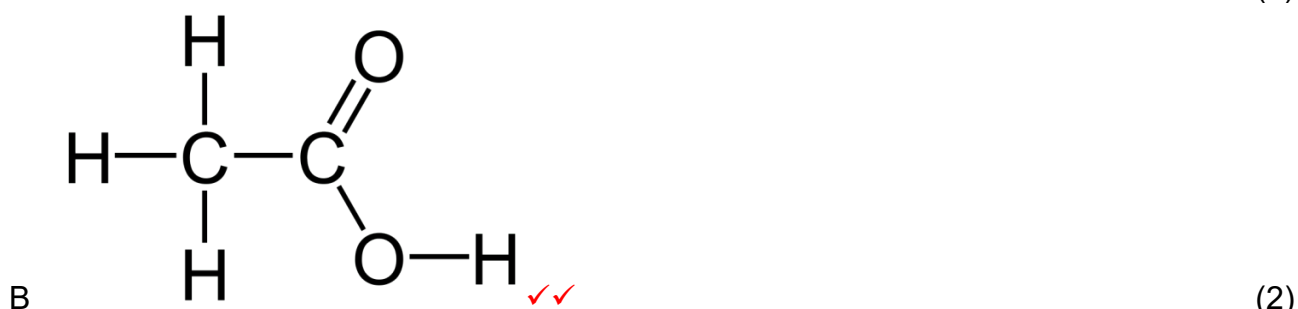
9.1.3 Carboxyl. ✓ (1)

9.1.4 Circle the ester✓ (1)

9.1.5 Carboxylic acid✓ (1)

9.1.6 Ester✓ (1)

9.1.7

9.1.8 $2 \text{C}_3\text{H}_8\text{O}$ or $\text{C}_3\text{H}_7\text{OH} + 9\text{O}_2 \rightarrow 6\text{CO}_2 + 8\text{H}_2\text{O}$ ✓ (3)

9.2

9.2.1 Octa✓-1,3-di✓ene✓ (3)

9.2.2 i is shorter ✓ that ii, as it has the additional of a pi bond✓, whereas ii, only has a sigma bond. ✓ (3)

9.2.3 Q

A The bromine molecule is red✓, and loses its colour✓ when bonded in the compounds✓. (3)

B Place equal amount of each compound in a test tube. ✓
Count the number of drops✓, until the colour stops disappearing. ✓
Compound A will take the most drops. ✓ (4)

9.2.4 $\text{CH}_2\text{CH}(\text{CH}_2)_5\text{CH}_3 + \text{Br}_2 \rightarrow \text{CH}_2\text{BrCHBr}(\text{CH}_2)_5\text{CH}_3$ (3)